APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – I

Introduction and Solution Methods

Syllabus (as given)

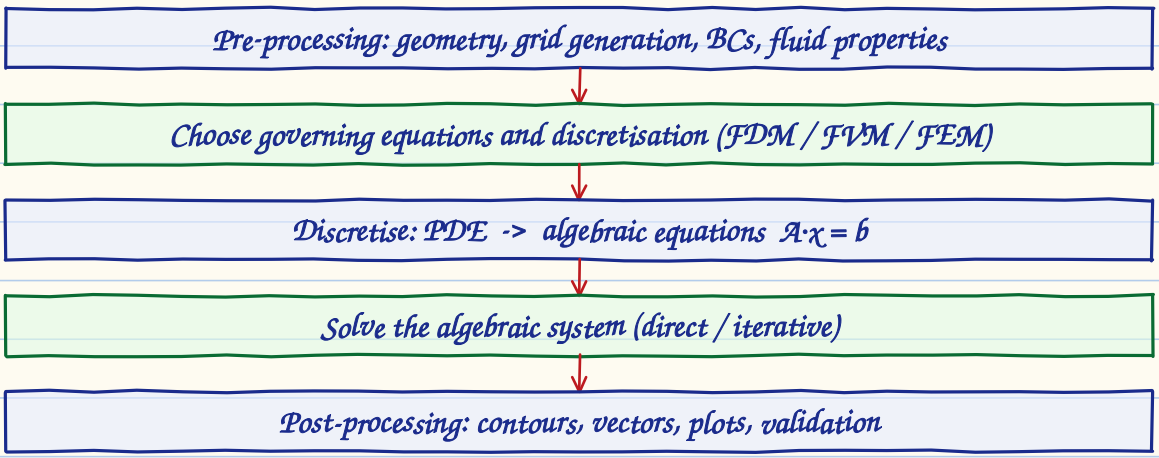
- Introduction: finite difference, finite volume and finite element methods, governing equations and boundary conditions, derivation of finite difference equations.
- Elliptic equations: finite difference formulations, iterative solution methods, direct method with Gaussian elimination.
- Parabolic equations: explicit schemes and Von Neumann stability analysis, implicit schemes, ADI, approximate factorization, fractional step methods, direct method with tridiagonal matrix algorithm (TDMA).

1. What is CFD?

Computational Fluid Dynamics = analysing fluid flow, heat transfer and related phenomena (combustion, chemical reactions) by solving the governing equations numerically on a computer. The continuous problem (PDEs) is converted into a discrete algebraic problem.

- Three tools of fluid dynamics: theory (analytical), experiment (wind tunnel), and CFD (numerical).
- Advantages: low cost, fast design cycles, full flow-field details, can simulate conditions that are impossible/dangerous to test.
- Limitations: needs modelling (turbulence, chemistry), numerical errors, results must be validated.

Steps in a CFD analysis



2. Governing Equations

All fluid-flow equations come from three conservation laws: conservation of mass, momentum (Newton's 2nd law) and energy (1st law of thermodynamics).

Continuity (mass)

$$\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0 \quad \text{i.e.} \quad \partial\rho/\partial t + \nabla\cdot(\rho V) = 0$$

For incompressible flow (ρ = constant): $\nabla\cdot V = 0$.

Momentum (Navier–Stokes), x-direction

$$\rho(\partial u/\partial t + u\partial u/\partial x + v\partial u/\partial y + w\partial u/\partial z) = -\partial p/\partial x + \mu\nabla^2 u + \rho f_x$$

Similar equations for y and z. Left side = inertia (unsteady + convective); right side = pressure force + viscous force + body force.

Energy

$$\rho c_p(\partial T/\partial t + V\cdot\nabla T) = k\nabla^2 T + \Phi + q$$

Φ = viscous dissipation, q = heat source. For a solid, or low-speed flow, this reduces to the heat equation $\partial T/\partial t = \alpha\nabla^2 T$ with $\alpha = k/\rho c_p$.

Generic transport equation (very important)

$$\partial\varphi/\partial t + \nabla\cdot(\rho V\varphi) = \nabla\cdot(\Gamma\nabla\varphi) + S_\varphi$$

- terms: unsteady + convection = diffusion + source
- $\varphi = 1$ gives continuity, $\varphi = u$ gives momentum, $\varphi = T$ gives energy. One discretisation method serves all equations.

3. Classification of PDEs

Second-order quasi-linear PDE: $A\varphi_{xx} + B\varphi_{xy} + C\varphi_{yy} + D = 0$. Type depends on the discriminant $B^2 - 4AC$.

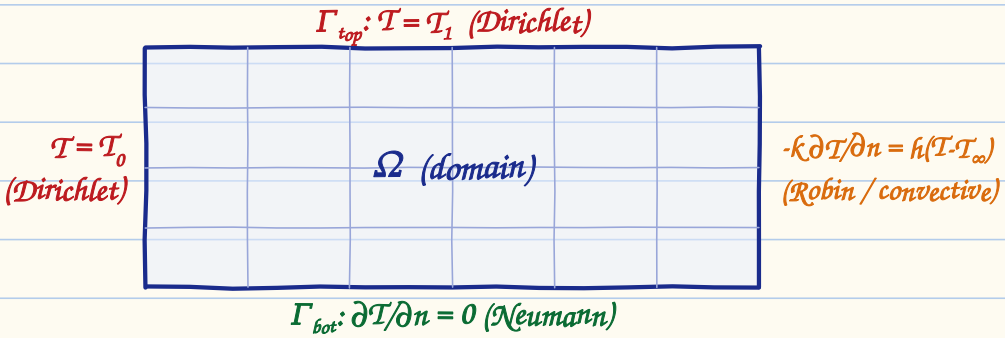
Type	B^2-4AC	Example	Physics
Elliptic	< 0	Laplace $\nabla^2\varphi=0$	steady, equilibrium
Parabolic	$= 0$	Heat $u_t=\alpha u_{xx}$	diffusion in time
Hyperbolic	> 0	Wave $u_{tt}=c^2u_{xx}$	wave propagation

- Elliptic: every point influences every other point; boundary conditions needed on the whole closed boundary -> boundary-value problem; solve all nodes together.
- Parabolic: information travels forward in time, spreads in space -> marching problem (initial +

boundary conditions).

- Hyperbolic: finite speed of propagation along characteristics -> marching with domain of dependence.

4. Boundary Conditions



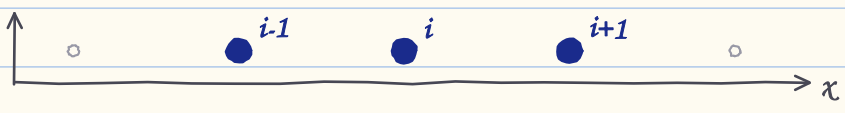
Type	Specified	Example
Dirichlet (1st)	value of φ	wall temperature, no-slip $u = 0$
Neumann (2nd)	normal derivative $\partial\varphi/\partial n$	insulated wall, symmetry plane
Robin (3rd)	$a\varphi + b\,\partial\varphi/\partial n = c$	convection $-kT_n = h(T - T_\infty)$

- A problem is well-posed if solution exists, is unique and depends continuously on data (Hadamard).
- Neumann conditions are implemented with a fictitious (ghost) node: $\partial\varphi/\partial x|_0 = 0$ -> $\varphi_{-1} = \varphi_1$.
- Initial condition $\varphi(x,0) = f(x)$ is needed for unsteady (parabolic / hyperbolic) problems.

5. Finite Difference Method (FDM)

Idea: replace the domain by a grid of discrete points and replace each derivative in the PDE by an algebraic difference of nodal values. The oldest and simplest method, best on structured grids.

1-D grid, spacing Δx



Derivation using Taylor series

$$u_{i+1} = u_i + \Delta x\, u_x + (\Delta x^2/2)\, u_{xx} + (\Delta x^3/6)\, u_{xxx} + (\Delta x^4/24)\, u_{xxxx} + \dots \quad (1)$$

$$u_{i-1} = u_i - \Delta x\, u_x + (\Delta x^2/2)\, u_{xx} - (\Delta x^3/6)\, u_{xxx} + (\Delta x^4/24)\, u_{xxxx} - \dots \quad (2)$$

(a) From (1): $u_x = (u_{i+1} - u_i)/\Delta x - (\Delta x/2)u_{xx} \dots$ Forward difference, $O(\Delta x)$

(b) From (2): $u_x = (u_i - u_{i-1})/\Delta x + (\Delta x/2)u_{xx} \dots$ Backward difference, $O(\Delta x)$

(c) (1) - (2): $u_x = (u_{i+1} - u_{i-1})/2\Delta x - (\Delta x^2/6)u_{xxx}$ Central difference, $O(\Delta x^2)$

(d) (1) + (2): $u_{xx} = (u_{i+1} - 2u_i + u_{i-1})/\Delta x^2 - (\Delta x^2/12)u_{xxxx}$ $O(\Delta x^2)$

The terms dropped are the truncation error; its leading power of Δx gives the order of accuracy.

Derivative	Formula	Order
u_x forward	$(u_{i+1} - u_i)/\Delta x$	1
u_x backward	$(u_i - u_{i-1})/\Delta x$	1
u_x central	$(u_{i+1} - u_{i-1})/(2\Delta x)$	2
u_{xx} central	$(u_{i+1} - 2u_i + u_{i-1})/\Delta x^2$	2
u_x 2nd-order upwind	$(3u_i - 4u_{i-1} + u_{i-2})/2\Delta x$	2

Mixed derivative: $u_{xy} \approx (u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}) / (4\Delta x\Delta y)$.

Concepts to remember

- Consistency: truncation error $\to 0$ as $\Delta x, \Delta t \to 0$.
- Stability: errors (round-off) do not grow as the solution marches.
- Convergence: numerical solution $\to$ exact solution as grid is refined.
- Lax equivalence theorem: for a well-posed linear initial-value problem, a consistent scheme is convergent if and only if it is stable.

6. Finite Volume and Finite Element Methods (overview)

Finite Volume Method (FVM)

- Domain divided into control volumes; the conservation law is integrated over each CV and the divergence theorem converts volume integrals into face fluxes.
- Fluxes leaving one cell = fluxes entering the neighbour -> conservative by construction. Works on unstructured grids. Basis of most commercial codes (Fluent, OpenFOAM, CFX).

Finite Element Method (FEM)

- Domain divided into elements (triangles / quadrilaterals); solution approximated as $u \approx \Sigma N_j(x)u_j$ with shape functions N_j; weighted-residual (Galerkin) or variational statement gives element matrices, then assembly.
- Excellent for complex geometry and natural boundary conditions; strong in structural and

Stokes-type flows (see Unit V).

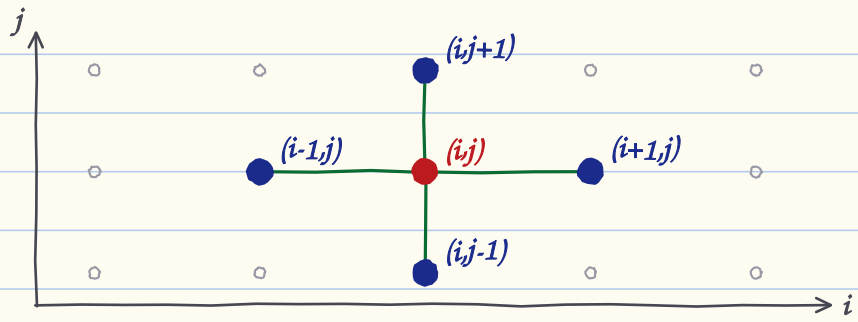
Feature	FDM	FVM	FEM
Basis	Taylor series	Integral balance	Weighted residual
Grid	structured	any	any (unstructured)
Conservation	not guaranteed	exact	weak sense
Complex geometry	difficult	good	very good

7. Solution Methods for Elliptic Equations

Model problem: Poisson equation $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = S(x,y)$ on a rectangle with φ given on the boundary (Dirichlet).

Finite-difference formulation (five-point formula)

Five-point stencil



Central differences in both directions with Δx, Δy:

$$(\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j})/\Delta x^2 + (\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1})/\Delta y^2 = S_{i,j}$$

With $\Delta x = \Delta y = h$ this gives the famous five-point formula:

$$\varphi_{i,j} = \tfrac{1}{4}\,(\varphi_{i+1,j} + \varphi_{i-1,j} + \varphi_{i,j+1} + \varphi_{i,j-1}) - (h^2/4)\,S_{i,j}$$

- Each interior node gives one linear equation; N interior nodes -> N equations -> system $A\varphi = b$ with a sparse, banded, diagonally dominant matrix.
- Value at a node = average of its four neighbours (Laplace equation) -> maximum principle: no interior maxima/minima.

8. Iterative (Interactive) Solution Methods

For large grids the matrix is huge but sparse; direct inversion is costly, so we start with a guess and improve it repeatedly. Write the five-point formula in iteration form ($S = 0$, $\Delta x = \Delta y$):

(i) Jacobi method

$$\varphi_{ij}^{(k+1)} = \tfrac{1}{4}\left(\varphi_{i+1,j}^{(k)} + \varphi_{i-1,j}^{(k)} + \varphi_{i,j+1}^{(k)} + \varphi_{i,j-1}^{(k)}\right)$$

Uses only old values; needs two arrays; slow.

(ii) Gauss–Seidel method

$$\varphi_{ij}^{(k+1)} = \tfrac{1}{4}\left(\varphi_{i+1,j}^{(k)} + \varphi_{i-1,j}^{(k+1)} + \varphi_{i,j+1}^{(k)} + \varphi_{i,j-1}^{(k+1)}\right)$$

Uses latest available values (sweeping i, j increasing); about twice as fast as Jacobi; one array only.

(iii) Successive Over-Relaxation (SOR)

$$\varphi^{(k+1)} = (1-\omega)\,\varphi^{(k)} + \omega\,\varphi_{GS}^{(k+1)}, \qquad 1 < \omega < 2$$

- Optimum factor for a square grid ($N \times N$): $\omega_{opt} \approx 2 / (1 + \sin(\pi/N))$.
- Line SOR / line Gauss–Seidel: solve a whole grid line implicitly with TDMA, sweep line by line (faster convergence).
- Convergence criterion: $\max |\varphi^{(k+1)} - \varphi^{(k)}| < \varepsilon$ (e.g. 10^{-5}), or residual norm $||b - A\varphi|| < \varepsilon$.
- Gauss–Seidel / Jacobi converge when the matrix is diagonally dominant (Scarborough: $\Sigma |a_{nb}| \leq |a_p|$, strict at least once).

Method	Iterations	Storage	Remark
Jacobi	~ N^2	2 arrays	parallel friendly
Gauss–Seidel	~ $N^2/2$	1 array	2× Jacobi
SOR (ω opt)	~ N	1 array	fast, ω critical

Worked example: Gauss–Seidel on a 2×2 interior grid

Plate: top edge = 100°, other three edges = 0°. Unknowns A, B (top row), C, D (bottom row).

$A = (100 + B + C)/4$, $B = (100 + A + D)/4$, $C = (A + D)/4$, $D = (B + C)/4$. Start $A = B = C = D = 0$.

Iteration 1: $A = 25$, $B = 31.25$, $C = 6.25$, $D = 9.375$

Iteration 2: $A = 34.375$, $B = 35.94$, $C = 10.94$, $D = 11.72$

Exact (by symmetry): $A = B = 37.5$, $C = D = 12.5$ -> converging.

9. Direct Method: Gaussian Elimination

Solves $A x = b$ exactly (up to round-off) in a finite number of operations.

1. Forward elimination: use row k to eliminate x_k from all rows below: $a_{ij} \leftarrow a_{ij} - (a_{ik}/a_{kk})\,a_{kj}$, $b_i \leftarrow$

$b_i - (a_{ik}/a_{kk})\, b_k$. The matrix becomes upper triangular $U\,x = c$.

2. **Back substitution**: $x_n = c_n/u_{nn}$, $x_i = (c_i - \Sigma_{j>i}\, u_{ij}x_j)/u_{ii}$.
3. Partial pivoting (swap rows so that the pivot is largest) prevents division by a tiny number.

- Operation count ≈ $(2/3)n^3$ for full matrix; for a banded matrix of half-bandwidth m only ≈ $n\,m^2$. For an $N\times N$ grid, $n = N^2$, $m = N$ -> N^4 operations — expensive, so iterative / fast solvers preferred for big 2-D, 3-D problems.
- Disadvantages: fill-in inside the band, large memory, round-off accumulation.

Remember

- Direct: exact, robust, cost grows rapidly with n.
- Iterative: cheap per iteration, needs convergence; best for large sparse systems.

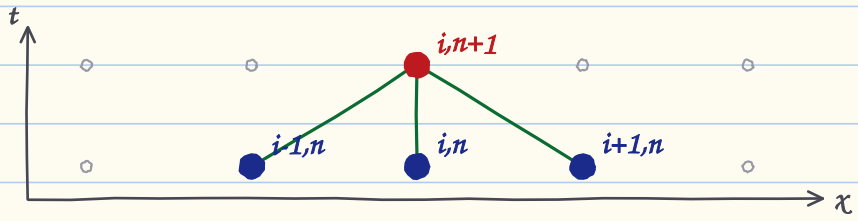
10. Parabolic Equations

Model: one-dimensional unsteady heat (diffusion) equation $\partial u/\partial t = \alpha\, \partial^2 u/\partial x^2$, $0 < x < L$, $t > 0$, with initial $u(x,0)$ and boundary values $u(0,t)$, $u(L,t)$.

Grid: $x_i = i\Delta x$, $t^n = n\Delta t$, $u_i^n \approx u(x_i, t^n)$. Diffusion number $r = \alpha\Delta t/\Delta x^2$.

Explicit scheme: FTCS (Forward-Time, Central-Space)

FTCS stencil: one unknown



$$(u_i^{n+1} - u_i^n)/\Delta t = \alpha\, (u_{i+1}^n - 2u_i^n + u_{i-1}^n)/\Delta x^2$$

$$u_i^{n+1} = r\, u_{i-1}^n + (1 - 2r)\, u_i^n + r\, u_{i+1}^n$$

- Truncation error $O(\Delta t) + O(\Delta x^2)$; each new value is found directly (no matrix) — cheap and simple.
- Condition for positive coefficients (physically realistic, no oscillations): $r \le ½$.

11. Von Neumann Stability Analysis

Method: represent the error (or solution) as a Fourier mode and see whether its amplitude grows or decays in one time step. Valid for linear, constant-coefficient schemes (periodic / ignoring boundaries).

Step 1 Substitute $u_i^n = g^n e^{I k x_i} = g^n e^{I i\theta}$, $\theta = k\Delta x$ ($I = \sqrt{-1}$).

Step 2 Cancel common factors -> amplification factor $G = g^{n+1}/g^n$.

Step 3 Stable if $|G| \le 1$ for all θ in $[0, \pi]$.

Apply to FTCS heat equation

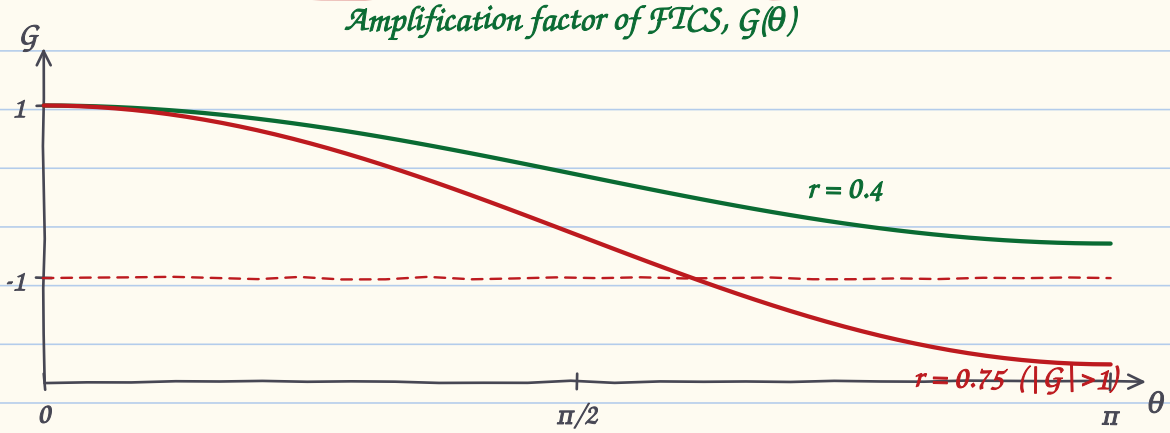
$$G\, e^{I\theta} = r\, e^{I(i-1)\theta} + (1-2r)\, e^{I\theta} + r\, e^{I(i+1)\theta}$$

$$G = 1 - 2r + 2r\cos\theta = 1 - 4r\sin^2(\theta/2)$$

G is real, $G \le 1$ always. Need $G \ge -1$: $1 - 4r \ge -1$ (worst case $\theta = \pi$) ->

$$r = \alpha\Delta t/\Delta x^2 \le 1/2 \quad \text{(conditionally stable)}$$

Amplification factor of FTCS, $G(\theta)$

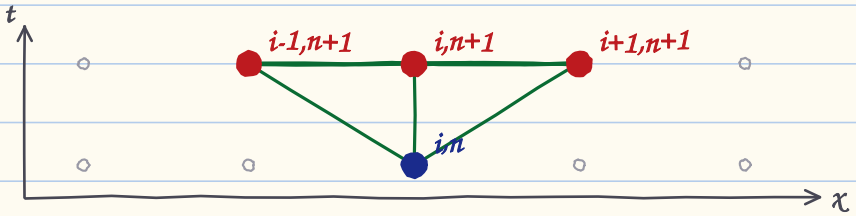


- Time step restriction $\Delta t \le \Delta x^2/(2\alpha)$: halving Δx forces Δt to be 4 times smaller -> explicit is costly on fine grids.

12. Implicit Schemes

Fully implicit (BTCS, Laasonen)

BTCS stencil: three unknowns



$$-r\, u_{i-1}^{n+1} + (1+2r)\, u_i^{n+1} - r\, u_{i+1}^{n+1} = u_i^n$$

- Gives a tridiagonal system at every time level (solved by TDMA). Von Neumann: $G = 1/(1 + 4r\sin^2(\theta/2))$ -> $|G| \le 1$ for all r: unconditionally stable.
- Accuracy $O(\Delta t) + O(\Delta x^2)$; Δt can be large, limited only by accuracy.

Crank–Nicolson (average of n and n+1)

$$-(r/2)u_{i-1}^{n+1} + (1+r)u_i^{n+1} - (r/2)u_{i+1}^{n+1} = (r/2)u_{i-1}^{n} + (1-r)u_i^{n} + (r/2)u_{i+1}^{n}$$

- Second order in time and space $O(\Delta t^2) + O(\Delta x^2)$; $G = (1 - 2r \sin^2(\theta/2))/(1 + 2r \sin^2(\theta/2))$; unconditionally stable; may oscillate for very large r.

Scheme	Order	Stability	Matrix
FTCS	$\Delta t, \Delta x^2$	$r \le 1/2$	none
BTCS	$\Delta t, \Delta x^2$	unconditional	tridiagonal
Crank–Nicolson	$\Delta t^2, \Delta x^2$	unconditional	tridiagonal
Richardson	$\Delta t^2, \Delta x^2$	unstable always	none
DuFort–Frankel	$\Delta t^2, \Delta x^2$	unconditional	none

Note: Richardson (leapfrog in time) is always unstable for diffusion; DuFort–Frankel replaces $2u_i^n$ by $(u_i^{n+1} + u_i^{n-1})$ and becomes stable but may be inconsistent unless $\Delta t/\Delta x \to 0$.

13. Direct Method: Tridiagonal Matrix Algorithm (Thomas)

Tridiagonal system $A \cdot T = d$

$$\begin{vmatrix} b_1 & c_1 & 0 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 0 \\ 0 & a_3 & b_3 & c_3 & 0 \\ 0 & 0 & a_4 & b_4 & c_4 \\ 0 & 0 & 0 & a_5 & b_5 \end{vmatrix} \quad [T_1\ T_2\ T_3\ T_4\ T_5] = [d_1\ d_2\ d_3\ d_4\ d_5]$$

b = diagonal, c = upper, a = lower

Equation i: $a_i T_{i-1} + b_i T_i + c_i T_{i+1} = d_i$. Gaussian elimination reduces to this simple two-sweep algorithm:

1. Forward sweep: $c'_1 = c_1/b_1$, $d'_1 = d_1/b_1$;

$$c'_i = c_i/(b_i - a_i c'_{i-1}), \quad d'_i = (d_i - a_i d'_{i-1})/(b_i - a_i c'_{i-1})$$

2. Back substitution: $T_n = d'_n$, $T_i = d'_i - c'_i T_{i+1}$ $(i = n-1, \ldots, 1)$.

- Only about 8n operations — O(n) instead of $O(n^3)$; needs diagonal dominance ($|b_i| > |a_i| + |c_i|$) for stability without pivoting.

Worked example

$2T_1 - T_2 = 1$; $-T_1 + 2T_2 - T_3 = 0$; $-T_2 + 2T_3 = 1$

$c'_1 = -1/2$, $d'_1 = 1/2$; $c'_2 = -1/(2 - 1/2) = -2/3$, $d'_2 = (0 + 1/2)/(3/2) = 1/3$; $d'_3 = (1 + 1/3)/(2 - 2/3) = 1$

Back substitution: $T_3 = 1$, $T_2 = 1/3 + (2/3)(1) = 1$, $T_1 = 1/2 + (1/2)(1) = 1$. Check: $2(1) - 1 = 1$ √

14. Multi-dimensional Parabolic Problems

2-D heat equation $u_t = \alpha(u_{xx} + u_{yy})$. Explicit FTCS needs $r_x + r_y \le ½$ (for $\Delta x = \Delta y$: $r \le ¼$). Fully implicit gives a pentadiagonal (5-point) matrix at each step — TDMA no longer applies directly.

Alternating Direction Implicit (ADI) — Peaceman–Rachford

Split each time step into two half steps; implicit in one direction, explicit in the other, so that only tridiagonal systems arise.

$$\text{Half step 1:}\quad (1 - (r_x/2)\delta_x^2)\, u^{n+1/2} = (1 + (r_y/2)\delta_y^2)\, u^n \qquad \text{(implicit in } x\text{)}$$

$$\text{Half step 2:}\quad (1 - (r_y/2)\delta_y^2)\, u^{n+1} = (1 + (r_x/2)\delta_x^2)\, u^{n+1/2} \qquad \text{(implicit in } y\text{)}$$

- $\delta_x^2 u = u_{i+1} - 2u_i + u_{i-1}$ (central second difference operator).
- Each half step = set of independent tridiagonal systems along grid lines -> TDMA. Second order $O(\Delta t^2, \Delta x^2)$, unconditionally stable in 2-D (not in 3-D with the simple extension; use Douglas–Gunn).

Approximate Factorization (Beam–Warming, delta form)

Implicit 2-D scheme: $(1 - r\theta(\delta_x^2 + \delta_y^2))\,\Delta u = r(\delta_x^2 + \delta_y^2)u^n$, $\Delta u = u^{n+1} - u^n$. Factor the operator:

$$(1 - r\theta\delta_x^2)(1 - r\theta\delta_y^2)\,\Delta u = r(\delta_x^2 + \delta_y^2)\,u^n$$

- The product introduces an extra term $r^2\theta^2\delta_x^2\delta_y^2\Delta u = O(\Delta t^3)$ — same order as the truncation error, so accuracy is preserved.
- Solve two sweeps: $(1 - r\theta\delta_x^2)\Delta u^* = RHS$ then $(1 - r\theta\delta_y^2)\Delta u = \Delta u^*$, each by TDMA; update $u^{n+1} = u^n + \Delta u$.

Fractional Step (Yanenko) Method

Split the multi-dimensional operator into 1-D operators and advance them successively:

$$(1 - r\delta_x^2)\, u^* = u^n \;; \qquad (1 - r\delta_y^2)\, u^{n+1} = u^*$$

- Each step is a 1-D implicit solve (TDMA). First order in time in general (split error); stable for any r. Easily extended to 3-D and to non-linear terms (each fractional step handles one physical

process: convection, diffusion, source).

15. Worked Example: Explicit Heat Conduction

A rod of length 1 m, $\alpha = 0.01$ m²/s, initial temperature 0, ends held at $u(0,t) = 0$ and $u(1,t) = 100$. Take $\Delta x = 0.25$ m (3 interior nodes) and $\Delta t = 1$ s.

1. Diffusion number $r = \alpha\Delta t/\Delta x^2 = 0.01(1)/0.0625 = 0.16 \le 0.5$ -> explicit scheme is stable.
2. Update formula: $u_i^{n+1} = 0.16\, u_{i-1}^n + 0.68\, u_i^n + 0.16\, u_{i+1}^n$.
3. Step 1 ($n = 0$ -> 1): $u_1 = 0.16(0) + 0.68(0) + 0.16(0) = 0$, $u_2 = 0$, $u_3 = 0.16(0) + 0.68(0) + 0.16(100) = 16$.
4. Step 2: $u_1 = 0$, $u_2 = 0.16(0) + 0.68(0) + 0.16(16) = 2.56$, $u_3 = 0.16(0) + 0.68(16) + 0.16(100) = 26.88$.

Heat diffuses in from the hot end one node per time step — a visible sign that explicit schemes carry information over only one cell per step.

16. Truncation Error, Modified Equation and Accuracy

Substituting Taylor expansions into FTCS gives the modified equation that the scheme really solves:

$$u_t = \alpha\, u_{xx} + (\Delta t/2)\, u_{tt} - (\alpha\Delta x^2/12)\, u_{xxxx} + \dots$$

- Leading error terms: $O(\Delta t)$ from the forward time difference and $O(\Delta x^2)$ from the central space difference.
- Choosing $r = 1/6$ ($\Delta t = \Delta x^2/6\alpha$) cancels the two leading terms and gives $O(\Delta t^2, \Delta x^4)$ accuracy for the explicit scheme — a useful trick.
- Grid refinement test: error ratio between grids Δx and $\Delta x/2$ is $\approx 2^p$; this gives the observed order p.
- Round-off errors are amplified when $|G| > 1$; discretisation errors are controlled by order + grid size. Stability is separate from accuracy.

Boundary treatment in the heat equation

- Dirichlet: simply use the given value for u_0^n and u_N^n.
- Neumann $\partial u/\partial x = q$ at $x = 0$: one-sided $u_x \approx (-3u_0 + 4u_1 - u_2)/2\Delta x = q$ (2nd order), or ghost node $u_{-1} = u_1 - 2\Delta x\, q$ (central).
- Insulated end ($q = 0$): $u_{-1} = u_1$, so $u_0^{n+1} = 2r\, u_1^n + (1-2r)\, u_0^n$.

Quick revision — Unit I

- Parabolic explicit FTCS: $r = \alpha\Delta t/\Delta x^2 \le 1/2$; BTCS and Crank–Nicolson: unconditionally stable.
- Von Neumann: put $u = g^n e^{I\theta}$, find G, need $|G| \le 1$.
- Elliptic: five-point formula; Jacobi < Gauss–Seidel < SOR.
- Direct: Gauss elimination $O(n^3)$; TDMA $O(n)$ for tridiagonal systems.
- Multi-D implicit: ADI, approximate factorization, fractional step — all reduce to repeated TDMA.

Important Questions

1. Derive central, forward and backward difference formulas using Taylor series and state their order.
2. Classify PDEs and give one example for each class.
3. Derive the five-point formula; compare Jacobi, Gauss–Seidel and SOR.
4. Explain Von Neumann stability analysis and find the stability limit of FTCS for the heat equation.
5. Explain Thomas algorithm with an example; explain ADI and fractional step methods.